

Computer systems 05/19

Registration No: 2025/CS1079

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(1)

$$\begin{array}{r} 1) \quad 11111 \\ + 11011 \\ \hline 111000_2 \end{array}$$

$$\begin{array}{r} 2) \quad 010001 \\ + 110000 \\ \hline 1000001_2 \end{array}$$

$$\begin{array}{r} 3) \quad 0110110 \\ + 1101110 \\ \hline 10100100_2 \end{array}$$

$$\begin{array}{r} 4) \quad 111000 \\ + 110000 \\ \hline 1101000_2 \end{array}$$

$$\begin{array}{r} 5) \quad 1110110 \\ + 0011111 \\ \hline 10010101_2 \end{array}$$

(2)

$$\begin{array}{r} 1) \quad 1111011 \\ - 1100011 \\ \hline 0011000_2 \end{array}$$

$$\begin{array}{r} 2) \quad 1100000 \\ - 0001100 \\ \hline 1010100_2 \end{array}$$

$$\begin{array}{r} 3) \quad 111111 \\ - 001111 \\ \hline 110000_2 \end{array}$$

$$\begin{array}{r} 4) \quad 1000000 \\ - 0011000 \\ \hline 0101000_2 \end{array}$$

$$\begin{array}{r} 5) \quad 1101101 \\ - 0001100 \\ \hline 1100001_2 \end{array}$$

3)

$$\begin{array}{r} 1) \quad 1011 \\ \quad \underline{111 \times} \\ \quad 1011 \\ \quad 10110 \\ \quad 101100 + \\ \hline 1001101_2 // \end{array}$$

$$\begin{array}{r} 2) \quad 100001 \\ \quad \underline{110 \times} \\ \quad 000000 \\ \quad 1000010 \\ \quad 10000100 + \\ \hline 11000110_2 // \end{array}$$

$$\begin{array}{r} 3) \quad 101010 \\ \quad \underline{110 \times} \\ \quad 000000 \\ \quad 1010100 \\ \quad 10101000 + \\ \hline 11111100_2 // \end{array}$$

$$\begin{array}{r} 4) \quad 111100 \\ \quad \underline{110 \times} \\ \quad 000000 \\ \quad 1111000 \\ \quad 11110000 + \\ \hline 101101000_2 // \end{array}$$

$$\begin{array}{r} 5) \quad 1010101 \\ \quad \underline{1010 \times} \\ \quad 0000000 \\ \quad 10101010 \\ \quad 000000000 \\ \quad 1010101000 + \\ \hline 1101010010_2 // \end{array}$$

4)

$$\begin{array}{r} 1) \quad -10 = 11110101_2 // \\ \quad 00001010_2 \\ \quad 11110101_2 \end{array}$$

$$\begin{array}{r} 2) \quad -23 = 11101000_2 // \\ \quad 00010111_2 \\ \quad 11101000_2 \end{array}$$

$$\begin{array}{r} 3) \quad -120 = 10000111_2 // \\ \quad 01111000_2 \\ \quad 10000111_2 \end{array}$$

$$\begin{array}{r} 4) \quad -32 = 11011111_2 // \\ \quad 00100000 \\ \quad 11011111 \end{array}$$

$$\begin{array}{r} 5) \quad -63 = 11000000_2 // \\ \quad 00111111_2 \\ \quad 11000000_2 \end{array}$$

5)

$$1) -10 = 1110110_2 //$$

$$00001010$$

$$1110101$$

$$\begin{array}{r} 1+ \\ \hline 1110110 \end{array}$$

$$2) -23 = 11101001_2 //$$

$$00010111$$

$$11101000$$

$$\begin{array}{r} 1+ \\ \hline 11101001 \end{array}$$

$$3) -120 = 10001000_2 //$$

$$01111000$$

$$10001111$$

$$\begin{array}{r} 1+ \\ \hline 10001000 \end{array}$$

$$4) -32 = 11100000_2 //$$

$$00100000$$

$$11011111$$

$$\begin{array}{r} +1 \\ \hline 11100000 \end{array}$$

$$5) -63 = 11000001_2 //$$

$$00111111$$

$$11000000$$

$$\begin{array}{r} 1+ \\ \hline 11000001 \end{array}$$

6)

$$1) 11.011_2 = 3.375 //$$

$$2+1.01\frac{1}{4}+\frac{1}{8}$$

$$3. \frac{3}{8}$$

$$2) 1101.11_2 = 13.75 //$$

$$8+4+1. \frac{1}{2}+\frac{1}{4}$$

$$13. \frac{3}{4}$$

$$3) 11011.101_2 = 27.625 //$$

$$16+8+2+1. \frac{1}{2}+\frac{1}{8}$$

$$27. \frac{5}{8}$$

$$4) 11.11_2 = 3.75 //$$

$$2+1. \frac{1}{2}+\frac{1}{4}$$

$$5) 101.001_2 = 5.125 //$$

$$4+1. \frac{1}{8}$$

⑦

1) - 256.2384

$256 = 1.00000000$

$0.2384 = 0.0011...$

$1.00000000.0011$

$1.000000000011 \times 2^8$

0.2384×2

0.4768×2

0.9536×2

1.8072×2

1.6144

biased exponent = $127 + 8$

$= 135$

$135 = 10000111_2$

$1 | 10000111 | 0000000000110000000000_2 //$

2) 234.1854

$234 = 11101010_2$

$0.1854 = 0.0010...$

$11101010.0010 \times$

1.11010100010×2^7

0.1854×2

0.3708×2

0.7416×2

1.4832×2

0.9664

biased exponent = $127 + 7$

$= 134$

$134 = 10000110_2$

$0 | 10000110 | 1101010001000000000000_2 //$

3) 324.23

$$324 = 101000100_2$$

$$0.23 = 0.0011$$

$$101000100.0011$$

$$1.010001000011 \times 2^8$$

$$0.23 \times 2$$

$$0.46 \times 2$$

$$0.92 \times 2$$

$$1.84 \times 2$$

$$1.68$$

$$\text{biased exponent} = 92 + 8$$

$$= 135$$

$$135 = 10000111_2$$

$$\boxed{0 \mid 10000111 \mid 010001000011000000000000} \quad 2//$$

4) -45.342

$$45 = 00101101_2$$

$$0.342 = 0.0101...$$

$$00101101.0101$$

$$1.011010101 \times 2^5$$

$$0.342 \times 2$$

$$0.684 \times 2$$

$$1.368 \times 2$$

$$0.736 \times 2$$

$$1.472$$

$$\text{biased exponent} = 127 + 5$$

$$= 132$$

$$132 = 10000100_2$$

$$\boxed{1 \mid 10000100 \mid 011010101000000000000000} \quad 2//$$

5) -432.245

$$432 = 110110000_2$$

$$0.245 = 0.0011 \dots$$

$$110110000.0011$$

$$1.101100000011 \times 2^8$$

$$0.245 \times 2$$

$$0.490 \times 2$$

$$0.980 \times 2$$

$$1.960 \times 2$$

$$1.920$$

$$\text{biased exponent} = 127 + 8$$

$$= 135$$

$$135 = 10000111_2$$

1	10000111	101100000011000000000000
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